

Lecture 04

Health Informatics

Microsoft Excel



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Excel

Content

- Excel Basics
- Formatting Cells and Worksheets
- Working with Data

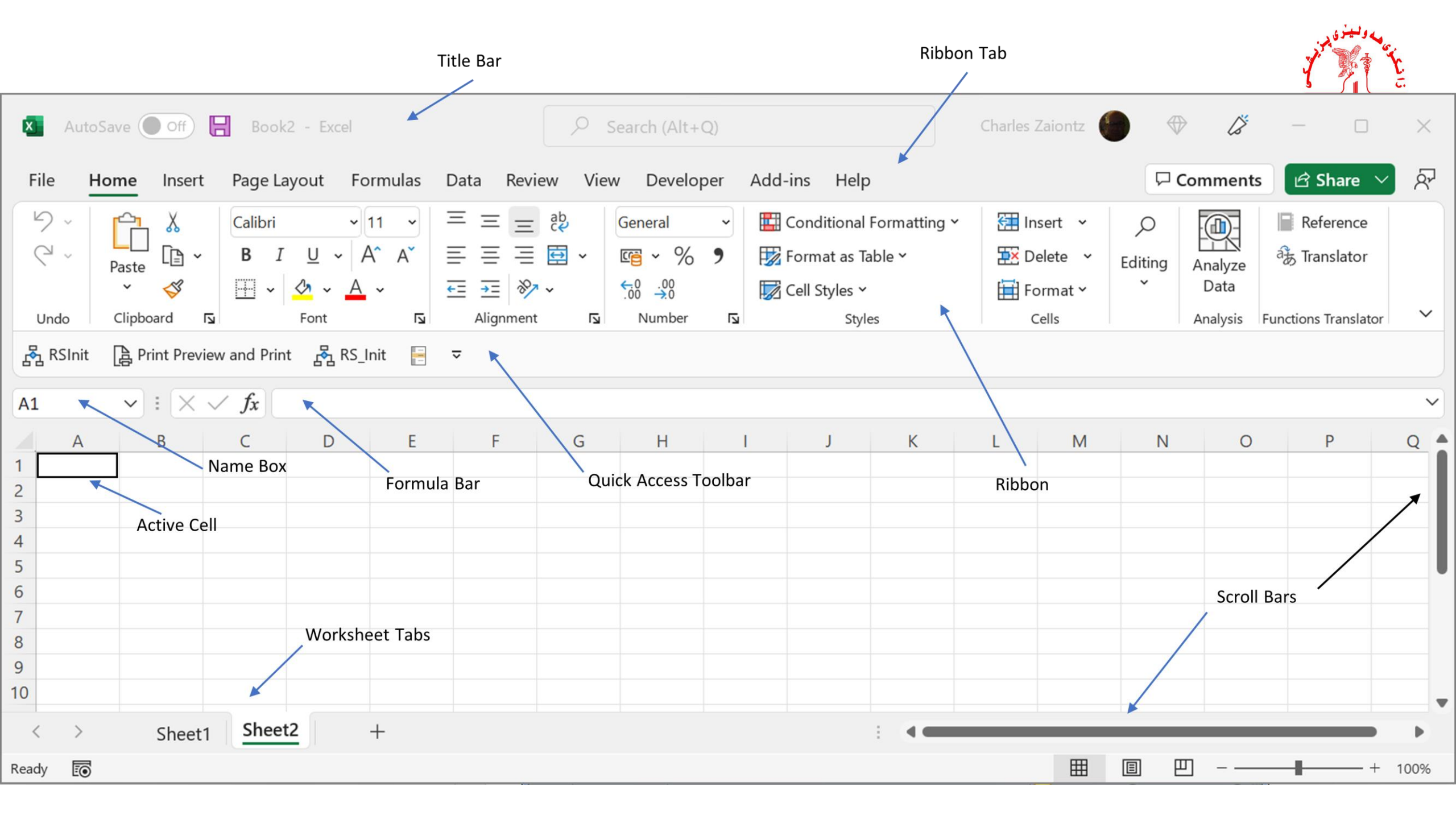


What is Microsoft Excel?

- Excel is a powerful spreadsheet software used for organizing, analyzing, and manipulating data.
- Common Uses:
 - Financial calculations
 - Data analysis
 - Budgeting
 - Project management

Excel Interface

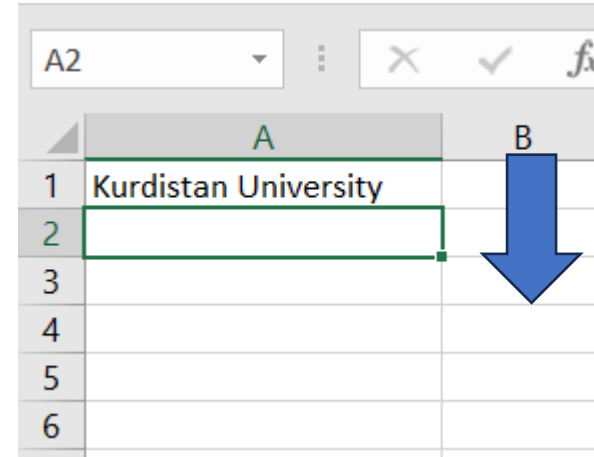
- Workbook: A collection of worksheets.
- Worksheet: A grid of cells for entering data.
- Cells: Intersections of rows and columns.
- Formulas: Expressions that perform calculations.



Entering Data

Steps:

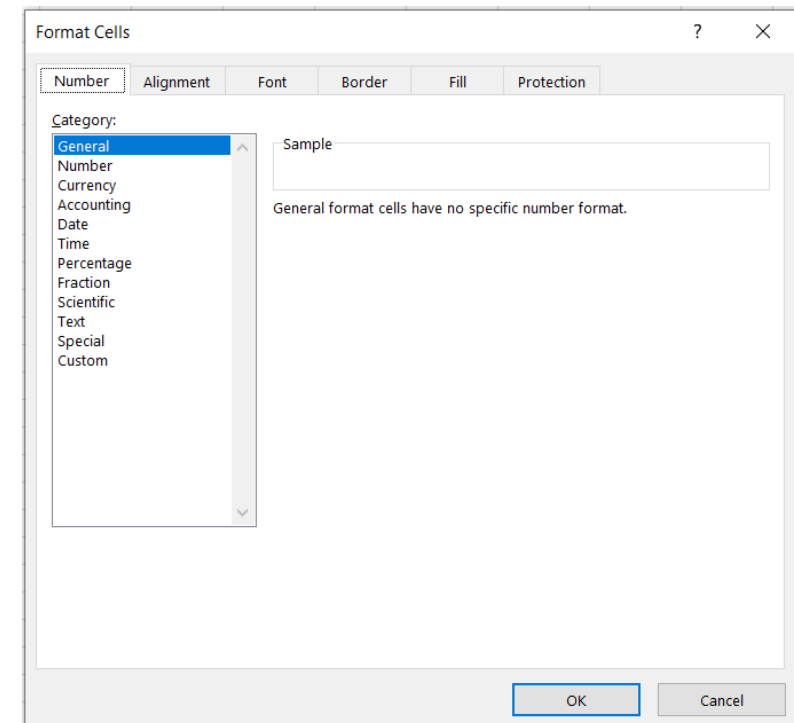
1. Click on a cell.
2. Type the data.
3. Press Enter to move to the next cell.



Formatting Cells

Options:

- Number format (currency, percentage, date, etc.)
- Alignment
- Font
- Fill color
- Borders



Using Formulas

Basic Operators:

- Addition (+)
- Subtraction (-)
- Multiplication (*)
- Division (/)

Simple Formulas

Example:

To calculate the sum of cells A1 and B1, enter " $=A1+B1$ " in a cell.

	A	B	C	D
1	2	4	$=A1+B1$	
2				
3				

Cell References

- Relative References: Adjust based on the cell's position.
- Absolute References: Remain fixed, even when copied.
- Use \$ before the row or column to make it absolute.

Common Functions

SUM: Adds a range of cells.

AVERAGE: Calculates the average of a range of cells.

COUNT: Counts the number of cells containing numbers.

Using Functions

Steps:

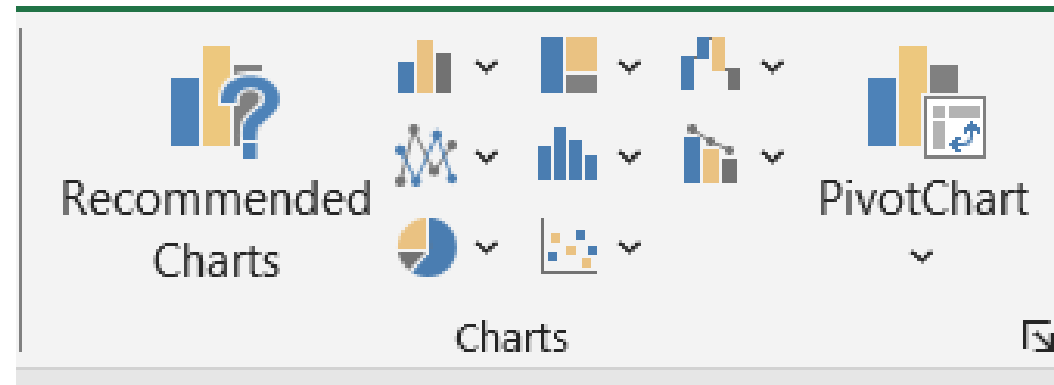
1. Type an equal sign (=).
2. Enter the function name.
3. Specify the range of cells.

	A	B	C	D	E
1	2				
2	5				
3	6				
4	7				
5	8				
6	8				
7	36				
8					

Creating Charts

Types of Charts:

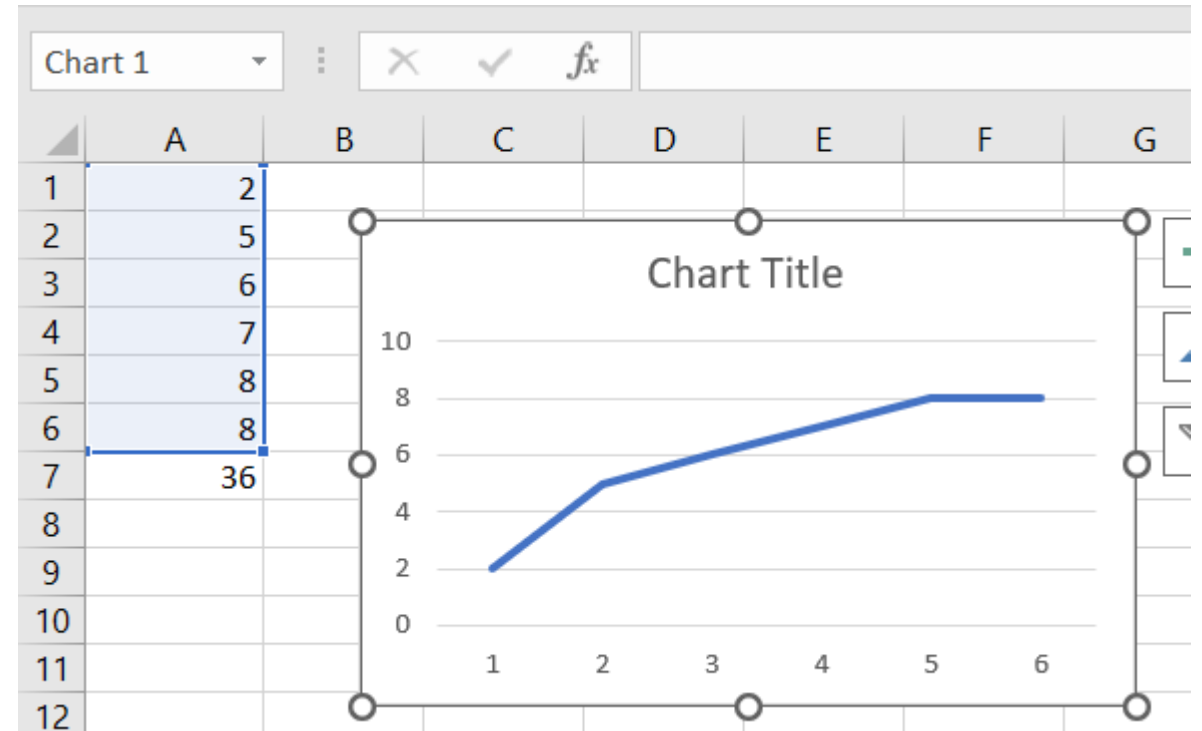
- Line charts
- Bar charts
- Pie charts
- Scatter plots



Creating a Chart

Steps:

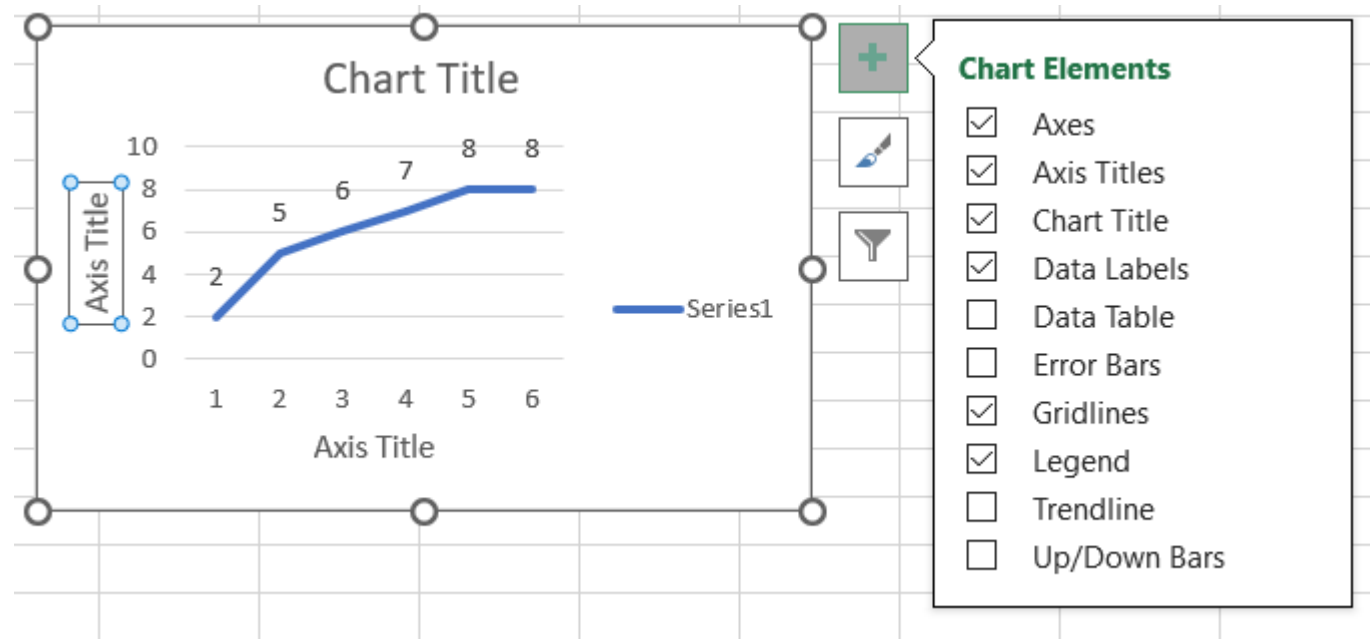
- Select the data range.
- Go to the "Insert" tab.
- Choose a chart type.



Customizing Charts

Options:

- Titles
- Labels
- Legends
- Colors



Sorting and Filtering Data

Sorting:

Arranging data in ascending or descending order.

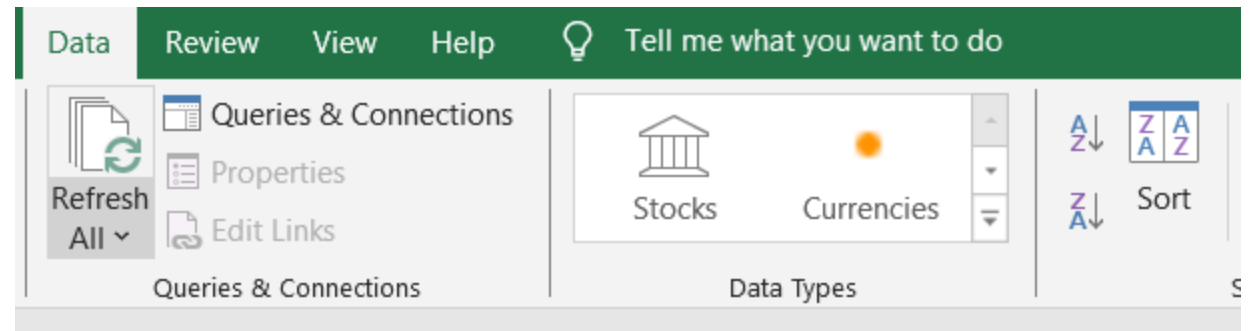
Filtering:

Displaying only specific data based on criteria.

Sorting Data

Steps:

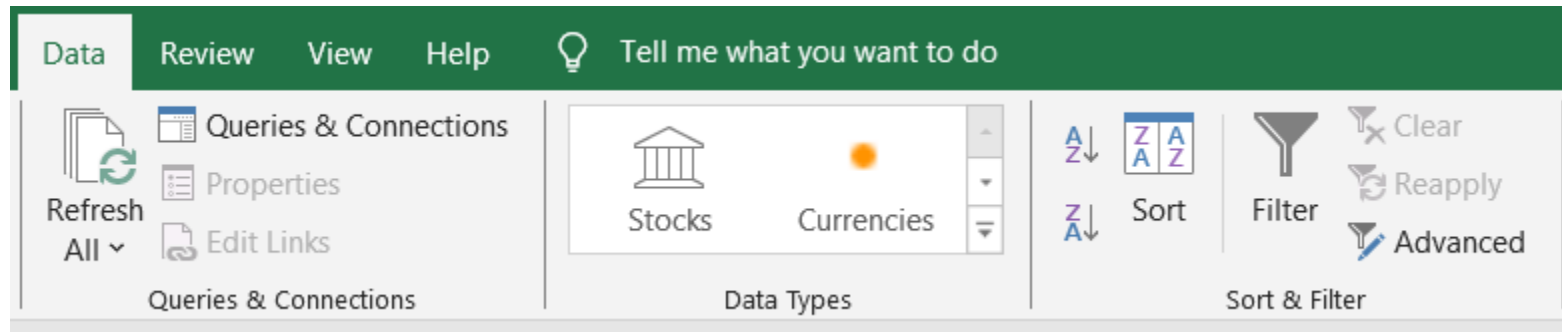
1. Select the data range.
2. Go to the "Data" tab.
3. Click "Sort."
4. Specify the sort criteria.



Filtering Data

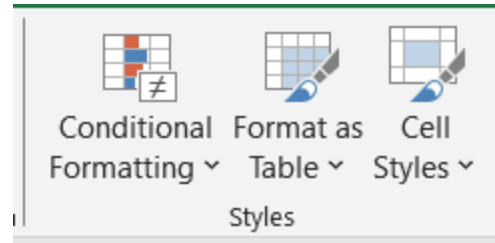
Steps:

1. Select the data range.
2. Go to the "Data" tab.
3. Click "Filter."
4. Use the filter drop-down menus to apply criteria.



Conditional Formatting

- Purpose: Apply formatting based on specific conditions.
- Options:
 - Highlight cells rules
 - Data bars
 - Color scales
 - Icon sets



Applying Conditional Formatting

Steps:

1. Select the data range.
2. Go to the "Home" tab.
3. Click "Conditional Formatting."
4. Choose a rule or create a custom rule.

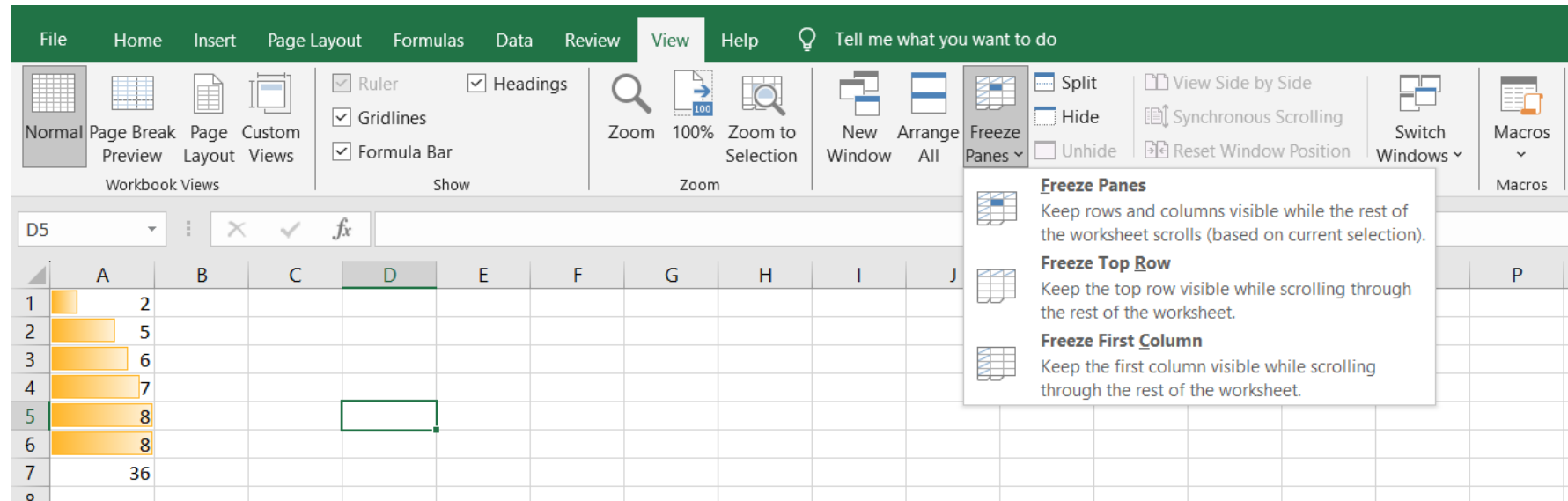
	A	B
1	2	
2	5	
3	6	
4	7	
5	8	
6	8	
7	36	

Freezing Panes

Purpose: Keep certain rows or columns visible while scrolling.

Steps:

1. Go to the "View" tab.
2. Click "Freeze Panes."

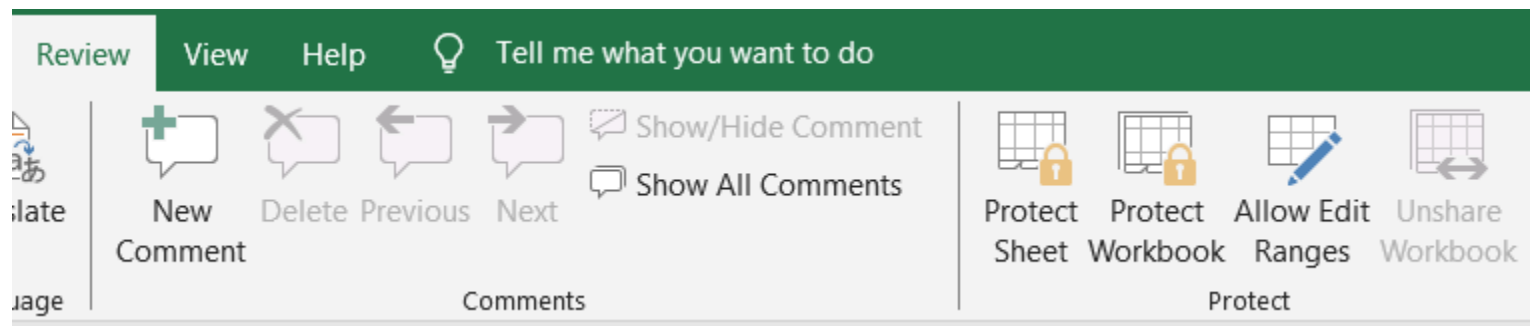


Protecting Worksheets

Purpose: Prevent unauthorized changes to data.

Steps:

1. Go to the "Review" tab.
2. Click "Protect Sheet."
3. Set a password.

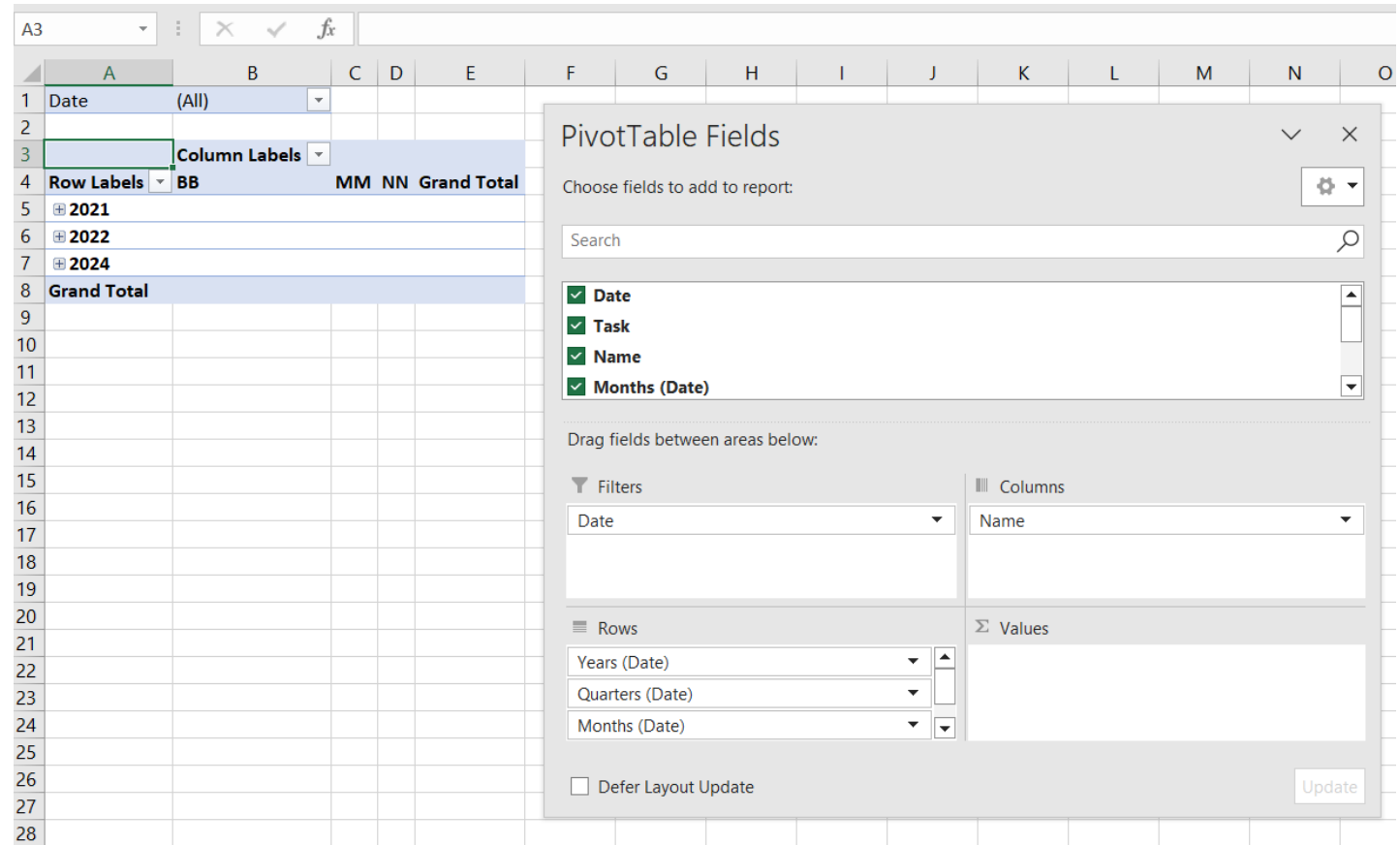


Creating Pivot Tables

Purpose: Summarize and analyze large datasets.

Steps:

1. Select the data range.
2. Go to the "Insert" tab.
3. Click "PivotTable."



The screenshot displays an Excel spreadsheet with a PivotTable and the PivotTable Fields task pane. The PivotTable is located in the range A3:O8 and has the following structure:

Date	(All)				
		Column Labels			
Row Labels	BB	MM	NN	Grand Total	
2021					
2022					
2024					
Grand Total					

The PivotTable Fields task pane is open on the right, showing the following configuration:

- Choose fields to add to report:** Date, Task, Name, Months (Date) are all checked.
- Drag fields between areas below:**
 - Filters:** Date
 - Columns:** Name
 - Rows:** Years (Date), Quarters (Date), Months (Date)
 - Values:** (Empty)
- Defer Layout Update:** ☐
- Update:** Button

Customizing Pivot Tables

Options:

Adding or removing fields.

Changing calculations.

Filtering data.

Exercise 1: Creating a Budget

Task: Create a budget worksheet with income and expenses.

Instructions:

1. Create a table with categories and amounts.
2. Use formulas to calculate totals.

Exercise 2: Analyzing Sales Data

Task: Create a chart to visualize sales trends.

Instructions:

1. Enter sales data into a table.
2. Create a line chart to show the trends.

Exercise 3: Using Conditional Formatting

Task: Highlight cells with sales above a certain threshold.

Instructions:

1. Apply conditional formatting to the sales data.
2. Set a rule to highlight cells above the threshold.



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